1b_10xPBMC_Stellarscope_celltype

August 31, 2026

[1]:
```
library(Seurat)
library(ggplot2)
library(pheatmap)
library(dplyr)
library(RColorBrewer)
library(reshape2)
library(clustree)
getwd()
dir.create("figures_10xPBMC_celltype")
dir.create("data_10xPBMC")
dataset_id <- "10xPBMC"
sample <- "pbmc8k"
mode <- "celltype"

colorPBMC <- "#81B29A"
```

```
Loading required package: SeuratObject

Loading required package: sp


Attaching package: ‘SeuratObject’


The following objects are masked from ‘package:base’:

    intersect, t



Attaching package: ‘dplyr’


The following objects are masked from ‘package:stats’:

    filter, lag
```

```
The following objects are masked from ‘package:base’:

    intersect, setdiff, setequal, union


Loading required package: ggraph


Attaching package: ‘ggraph’


The following object is masked from ‘package:sp’:

    geometry

```

'/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'

```
Warning message in dir.create("figures_10xPBMC_celltype"):
"'figures_10xPBMC_celltype' already exists"
Warning message in dir.create("data_10xPBMC"):
"'data_10xPBMC' already exists"
```

[2]:
```
# celltype annotation

celltypes <- read.table("/whitelists/celltype_annotation_sub_min.tsv") # same␣
 ↪file passed to snakemake

celltypes$V2 <- gsub(pattern="CD14pos_Monocytes", replacement =␣
 ↪"Monocytes_CD14pos", celltypes$V2)
celltypes$V2 <- gsub(pattern="FCGR3Apos_Monocytes", replacement =␣
 ↪"Monocytes_FCGR3Apos", celltypes$V2)
celltypes$V2 <- gsub(pattern="Naive_CD4_T", replacement = "T_CD4_Naive",␣
 ↪celltypes$V2)
celltypes$V2 <- gsub(pattern="Memory_CD4_T", replacement = "T_CD4_Memory",␣
 ↪celltypes$V2)
celltypes$V2 <- gsub(pattern="CD8A", replacement = "T_CD8", celltypes$V2)
celltypes$V2 <- gsub(pattern="Megakaryocytes", replacement = "Platelet",␣
 ↪celltypes$V2)

PBMCcelltypeColors <- c("B"="#91bcca",
                "Monocytes_CD14pos"="#3d405b",
                "T_CD8"="#f2cc8f",
                "Dendritic"="#c492a7",
                "Monocytes_FCGR3Apos"="#e78063",
               # "Platelet"="#c4b3ab",
                "T_CD4_Memory"="#81b29a",
```

```
                "T_CD4_Naive"="#4d7362",
                "NK"= "#88535a")
```

[3]:
```
path <- paste0("/mnt/TEresults2/snakemake_results/results/stellarscope_out/",␣
↪dataset_id, "/", sample, "/", mode, "/")


TEmatrix <- Seurat::ReadMtx(mtx = paste0(path, sample, "_", mode, "-TE_counts.
↪mtx"),
                             cells = paste0(path, sample, "_", mode,␣
↪"-barcodes.tsv"),
                             features = paste0(path, sample, "_", mode,␣
↪"-features.tsv"),
                             feature.column = 1) # read matrix
TEmatrix <- TEmatrix[setdiff(rownames(TEmatrix),"__no_feature"),]
nCells <- ncol(TEmatrix)
thrMinCells <- round(nCells * 0.05)

# create Seurat object with shallow filtering of TEs expressed in at least 50␣
↪cells and cells expressing at least 50 TEs
objTE_allCB <- Seurat::CreateSeuratObject(TEmatrix, project = paste0("PBMC_",␣
↪mode),
                         min.cells = thrMinCells, min.features = 50)

STAR_path <- paste0("/mnt/TEresults/snakemake_results/results/STARoutdir/",␣
↪dataset_id, "/", sample, "/best_Solo.out/Gene")
filteredBarcodes <- read.table(paste0(STAR_path, "/filtered/barcodes.tsv"))$V1␣
↪# read barcodes seleced by STARsolo


# barcodes of platelets
plateletBarcodes <- celltypes[celltypes$V2=="Platelet","V1"]
# remove low quality cells and platelets
objTE_stellarscope <- objTE_allCB[,setdiff(filteredBarcodes, plateletBarcodes)]␣
↪# keep only filtered barcodes

objTE_stellarscope
```

```
Warning message:
"Feature names cannot have underscores ('_'), replacing with dashes ('-')"
```

```
An object of class Seurat
6236 features across 1056 samples within 1 assay
Active assay: RNA (6236 features, 0 variable features)
 1 layer present: counts
```

[4]:
```
annotation_stellarscope <- read.table("annotation/annotation_stellarscope_hg38.
↪tsv") # generated with annotation_scripts/create_annotations_hg38.Rmd
head(annotation_stellarscope)
```

A data.frame: 6 × 10

	chr <chr>	start <int>	end <int>	sw_score <int>	strand <chr>	locusID <chr>	family <chr>	subfamily <chr>
1	chr1	67108754	67109046	1892	+	L1P5	L1	L1P5
2	chr1	8388316	8388618	2582	-	AluY	Alu	AluY
3	chr1	25165804	25166380	4085	+	L1MB5	L1	L1MB5
4	chr1	33554186	33554483	2285	-	AluSc	Alu	AluSc
5	chr1	41942895	41943205	2451	-	AluY_dup1	Alu	AluY
6	chr1	50331337	50332274	1587	+	HAL1	L1	HAL1

[5]:
```
# Remove some classes of TEs

classesToExclude <- c("Other", "Satellite", "Unknown", "RNA")

stellarscopeTEs <- Features(objTE_stellarscope)

TEsToKeep <- annotation_stellarscope[! annotation_stellarscope$class %in%␣
↪classesToExclude, ]$stellarscopeID

length(setdiff(stellarscopeTEs, TEsToKeep))/ length(stellarscopeTEs) * 100 #␣
↪percentage removed

objTE_stellarscope <- objTE_stellarscope[intersect(stellarscopeTEs, TEsToKeep),]
```

0.0160359204618345

[6]:
```
feature_metadata <- annotation_stellarscope[match(Features(objTE_stellarscope),␣
↪annotation_stellarscope$stellarscopeID),]

head(feature_metadata)
```

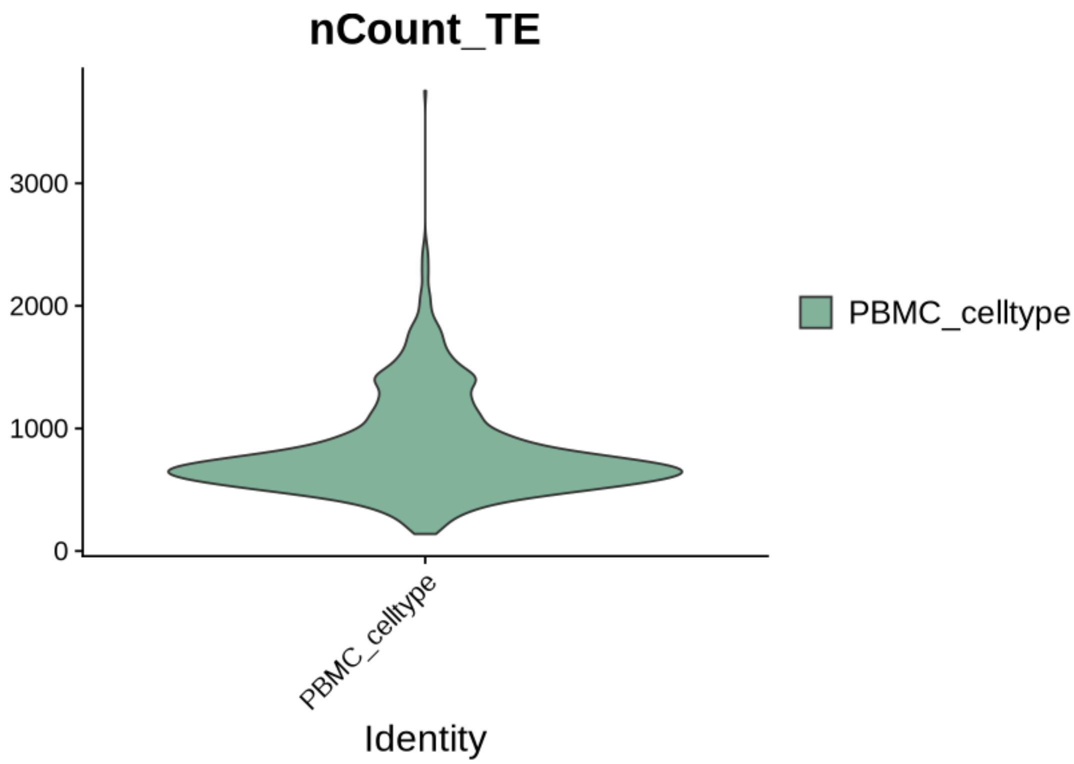
A data.frame: 6 × 10

	chr <chr>	start <int>	end <int>	sw_score <int>	strand <chr>	locusID <chr>	family <chr>	s <
269	chr1	43384732	43384958	1005	-	MIRb_dup9	MIR	M
584	chr1	150732607	150732908	2155	-	AluSz6_dup5	Alu	A
2217	chr1	958244	958549	2303	-	AluSx1_dup51	Alu	A
2515	chr1	1271693	1272004	2458	+	AluSp_dup39	Alu	A
2624	chr1	1394468	1394776	2377	+	AluSx_dup60	Alu	A
2752	chr1	1473369	1473495	1130	-	AluY_dup81	Alu	A

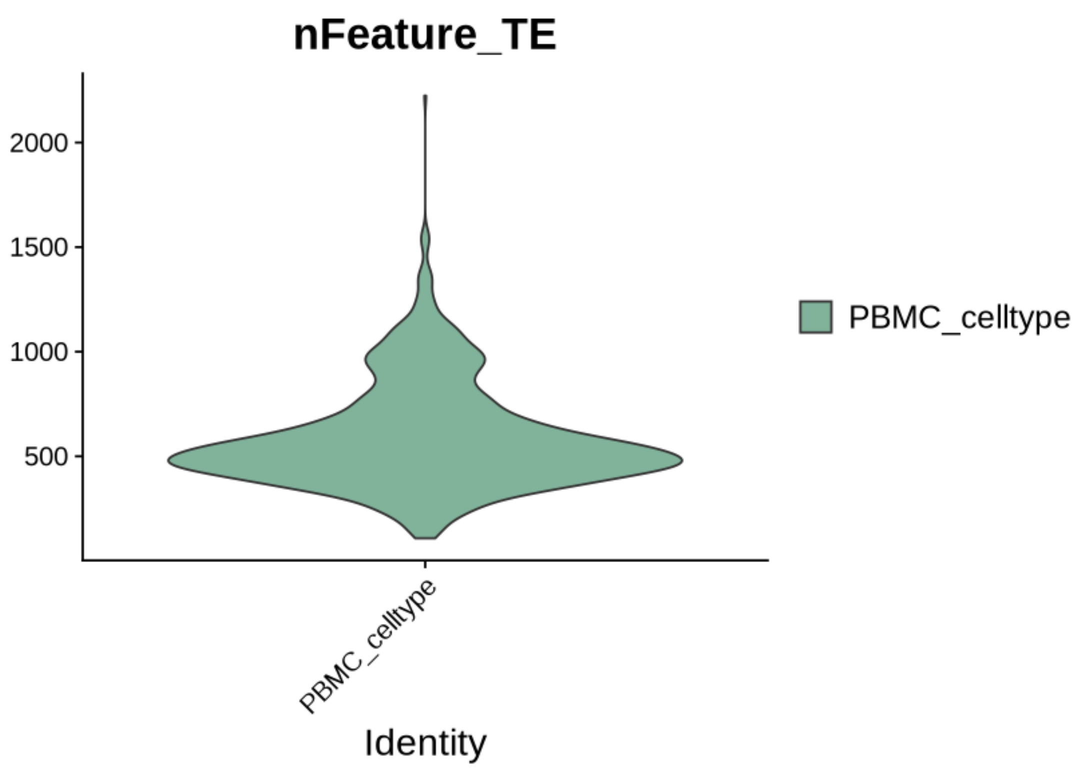
[7]:
```
options(repr.plot.width=7, repr.plot.height=5)


objTE_stellarscope@meta.data$nCount_TE <- objTE_stellarscope@meta.
↪data$nCount_RNA
```

```r
objTE_stellarscope@meta.data$nFeature_TE <- objTE_stellarscope@meta.
↪data$nFeature_RNA
# Visualize QC metrics as a violin plot
VlnPlot(objTE_stellarscope, features = c("nCount_TE"), ncol = 1,
        cols = colorPBMC, pt.size = 0) + theme(text=element_text(size=17))

ggsave(paste0("figures_",dataset_id,"/nCountTE_violin_stellarscope_",mode,".
↪png"), device='png',dpi=600)
ggsave(paste0("figures_",dataset_id,"/nCountTE_violin_stellarscope_",mode,".
↪pdf"), device='pdf')

VlnPlot(objTE_stellarscope, features = c("nFeature_TE"), ncol = 1,
        cols = colorPBMC, pt.size = 0) + theme(text=element_text(size=17))
ggsave(paste0("figures_",dataset_id,"/nFeatureTE_violin_stellarscope_",mode,".
↪png"), device='png',dpi=600)
ggsave(paste0("figures_",dataset_id,"/nFeatureTE_violin_stellarscope_",mode,".
↪pdf"), device='pdf')
```

```
Warning message:
"Default search for "data" layer in "RNA" assay yielded no results; utilizing
"counts" layer instead."

Warning message:
"The `slot` argument of `FetchData()` is deprecated as of SeuratObject
5.0.0.
  Please use the `layer` argument instead.
  The deprecated feature was likely used in the Seurat
package.
   Please report the issue at
<https://github.com/satijalab/seurat/issues>."
Warning message:
"`PackageCheck()` was deprecated in SeuratObject 5.0.0.
  Please use `rlang::check_installed()` instead.
  The deprecated feature was likely used in the Seurat
package.
   Please report the issue at
<https://github.com/satijalab/seurat/issues>."
Saving 7 x 7 in image
Saving 7 x 7 in image
Warning message:
"Default search for "data" layer in "RNA" assay yielded no results; utilizing
"counts" layer instead."
```



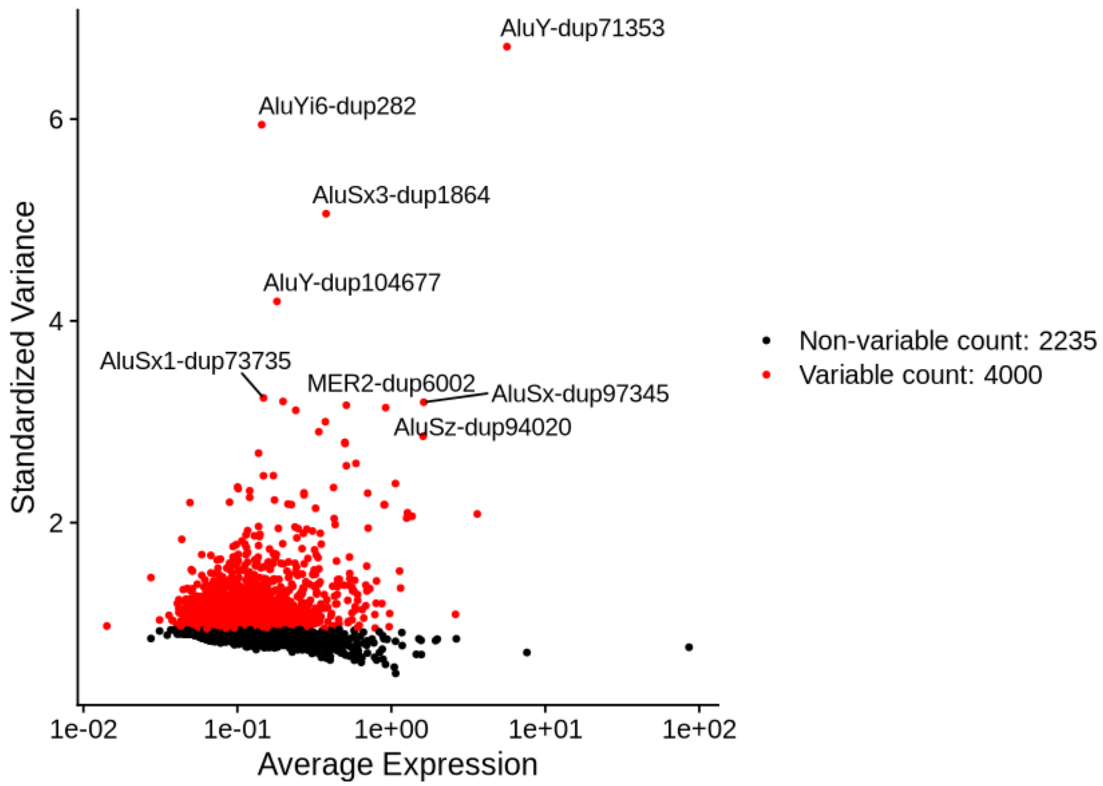

```
Saving 7 x 7 in image
Saving 7 x 7 in image
```



[8]:
```
objTE_stellarscope <- JoinLayers(objTE_stellarscope)

objTE_stellarscope <- NormalizeData(objTE_stellarscope, normalization.method =
 ↪"LogNormalize", scale.factor = 10000)

objTE_stellarscope <- FindVariableFeatures(objTE_stellarscope, selection.method
 ↪= "vst", nfeatures = 4000)

# Identify the 10 most highly variable genes
top10 <- head(VariableFeatures(objTE_stellarscope), 10)

# plot variable features with and without labels
plot1 <- VariableFeaturePlot(objTE_stellarscope)
plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
plot2
```

```
Normalizing layer: counts

Finding variable features for layer counts

When using repel, set xnudge and ynudge to 0 for optimal results
```

```
Warning message:
"ggrepel: 1 unlabeled data points (too many overlaps). Consider increasing
max.overlaps"
```



[9]:
```
gc()
all.genes <- rownames(objTE_stellarscope)
objTE_stellarscope <- ScaleData(objTE_stellarscope) # on hvgs
```

A matrix: 2 × 6 of type dbl

	used	(Mb)	gc trigger	(Mb)	max used	(Mb)
Ncells	18101848	966.8	45799250	2446.0	29123924	1555.4
Vcells	111596560	851.5	245602162	1873.8	245435577	1872.6

```
Centering and scaling data matrix
```

[10]:
```
objTE_stellarscope <- RunPCA(objTE_stellarscope, features =␣
↪VariableFeatures(object = objTE_stellarscope))

DimPlot(objTE_stellarscope, reduction = "pca") + NoLegend()

ElbowPlot(objTE_stellarscope)
```

```
PC_ 1
```

Positive: AluSz6-dup5, AluSz-dup85172, MIR3-dup81421, AluSz-dup94020, AluSc5-dup6172, L1ME3A-dup871, L2c-dup130701, FLAM-A-dup6099, AluY-dup71353, Tigger1-dup5817

AluJo-dup47192, AluJb-dup126028, MER5C1-dup277, MER39-dup143, AluSx1-dup84605, AluYa5-dup1820, MamRTE1-dup5254, AluJb-dup42076, AluJb-dup3312, MIR-dup162327

AluSq-dup11392, AluSc8-dup11182, AluY-dup30565, L1ME3Cz-dup8254, AluSz6-dup7653, L2a-dup163990, HAL1-dup18700, AluSp-dup29416, AluSp-dup32375, AluSz6-dup29860

Negative: L2b-dup1981, L2b-dup85510, AluSx-dup97345, AluSz-dup65970, Harlequin-int-dup426, AluJb-dup73017, MLT-int-dup273, AluSz-dup65021, AluSx-dup34093, AluSx1-dup96779

MLT1A-dup5928, AluSc8-dup7305, AluSp-dup3380, AluSc8-dup22443, AluSz-dup75971, AluJb-dup29720, AluSp-dup36543, AluSx3-dup21076, MIRc-dup98292, AluSx1-dup56311

AluSq2-dup15156, L1PA15-dup2196, MLT1A0-dup16914, AluSz-dup76216, L1M4-dup14436, AluJo-dup55958, MIRb-dup176657, MER92-int-dup473, L1MA9-dup11994, AluY-dup106656

PC_ 2

Positive: AluJr-dup89057, L1M5-dup2808, AluY-dup104962, AluSc5-dup6699, L1MC5a-dup13849, AluJr-dup84944, AluJr-dup5050, AluSx1-dup105649, AluJr-dup84945, AluSz-dup93905

MIRc-dup14097, AluJr-dup26147, AluSx1-dup86532, AluJb-dup43713, ORSL-dup906, MIRc-dup97273, MLT1A0-dup938, AluY-dup46259, Tigger9a-dup49, AluJb-dup86231

AluSc-dup20728, L1M5-dup40421, AluSq2-dup14424, AluSq-dup1994, AluSx1-dup13093, AluSz-dup40595, AluSx1-dup33189, AluY-dup61306, AluY-dup12183, L1MB5-dup1608

Negative: AluSx-dup97345, AluSz6-dup29860, AluJo-dup47192, AluJo-dup69143, AluY-dup30565, AluY-dup32753, AluY-dup71353, AluSx1-dup45337, Tigger1-dup5817, L2b-dup85510

AluSx1-dup107728, AluY-dup99592, AluSz-dup85172, FLAM-C-dup14640, AluSz-dup30155, L2a-dup163990, L1ME3A-dup8534, AluY-dup106670, AluSp-dup32375, MER5C1-dup277

L2a-dup163989, AluSg-dup22148, FLAM-C-dup19077, AluSx3-dup20249, AluSq2-dup8938, AluYe5-dup191, AluSz-dup65970, AluSz-dup84914, MIRc-dup98292, L2b-dup1981

PC_ 3

Positive: L2b-dup85510, L2b-dup1981, AluSq-dup8331, AluSx1-dup46526, AluSx1-dup39707, AluSx4-dup3994, AluSx1-dup88383, AluSx1-dup46199, AluY-dup44374, AluSp-dup36543

AluSx1-dup109068, MLT1A0-dup19532, AluSx1-dup51013, AluSc8-dup16056, AluJo-dup73332, AluSg-dup15181, AluSx-dup100843, L1PA2-dup2086, AluSz-dup76216, AluSc5-dup2592

AluSz-dup41379, AluJb-dup8078, MLT-int-dup273, AluY-dup98819, AluSz-dup65021, AluSz-dup73882, AluSx-dup70127, AluJr4-dup8919, Charlie4a-dup41, AluJr-dup4539

Negative: L1MC5a-dup13849, AluJr-dup89057, AluSc5-dup6699, AluY-dup104962, AluJr-dup84944, AluJr-dup84945, AluSx1-dup105649, AluSz-dup93905, ORSL-dup906, L1M5-dup2808

AluSx1-dup86532, AluJo-dup69143, AluJr-dup5050, AluSx-dup6825, MLT1A0-dup938, MIRc-dup97273, AluY-dup30565, AluSz6-dup5, AluJb-dup86231, MIRb-dup204693

AluSq2-dup14424, Tigger9a-dup49, AluSq-dup1899, L1M5-dup40421, AluSq-dup1994, MER39-dup143, L1MB5-dup1608, AluSx1-dup103159, AluSx-dup27178, AluSz6-dup32784

PC_ 4

Positive: AluY-dup30565, AluY-dup106670, AluY-dup71353, AluJr-dup72756, AluY-dup106766, AluSz-dup73882, AluY-dup43551, AluY-dup82338, AluSx1-dup73735, AluJb-dup3312

AluYe5-dup184, MIR-dup162327, AluSz-dup40595, MIRb-dup81134, AluSz-dup87950, AluSq2-dup3055, AluSx1-dup83623, AluSx-dup3431, AluSx1-dup30337, MER11C-dup315

AluSp-dup32315, AluSq2-dup58238, MIRb-dup140359, MIRc-dup2041, AluYm1-dup801, AluSx-dup103749, AluSx3-dup30890, AluY-dup15214, L2c-dup91716, L1PA10-dup1255

Negative: L3-dup2815, AluY-dup7033, AluJr-dup82212, AluSz6-dup22130, AluSc8-dup22494, AluJb-dup45602, AluSg-dup30170, L2a-dup163989, AluSc-dup32567, AluSc8-dup7305

MER41A-dup2380, AluY-dup13774, L1MA9-dup11533, L3-dup40489, L2a-dup163990, AluSz-dup84914, AluSx3-dup20134, HAL1-dup18700, AluJb-dup45601, L1PA4-dup5055

AluSx1-dup84603, AluSx1-dup84605, AluSz6-dup44790, AluSc5-dup6328, AluSx-dup47171, MER5A-dup24456, MIRc-dup98292, AluSx-dup99692, L2c-dup71375, AluSq-dup8146

PC_ 5

Positive: AluSc8-dup7305, AluSx-dup97345, MIRc-dup98292, AluSx1-dup65945, Tigger3b-dup268, MER30-dup2626, AluSz-dup65968, AluSx-dup44855, AluJo-dup61959, AluSx1-dup73638

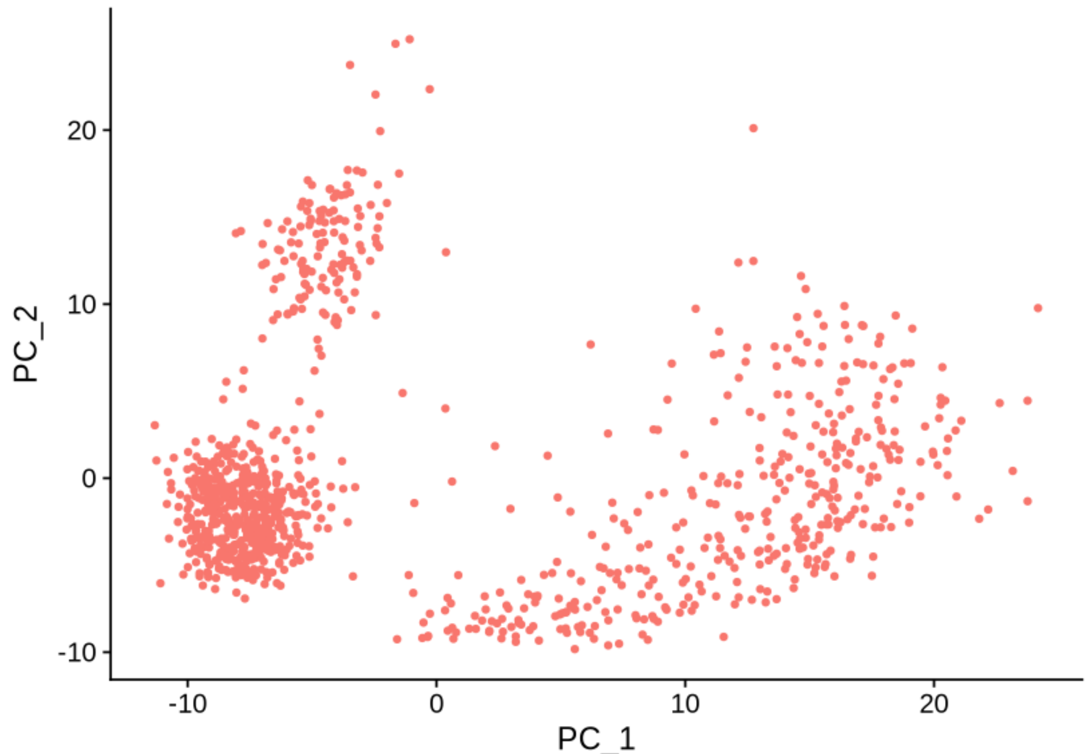
AluSq2-dup24157, L2a-dup142356, AluJb-dup5937, MLT-int-dup273, AluSx-dup5374, L2c-dup100575, AluSz-dup36471, AluSz-dup65970, AluSx1-dup84522, AluSx1-dup98432

AluY-dup70848, AluSz-dup73882, AluSx1-dup88436, L1MB7-dup15579, L2a-dup77267, AluY-dup99592, AluY-dup106766, AluSq2-dup3055, AluJr-dup33220, AluSp-dup37782

Negative: AluSc8-dup22494, AluSc8-dup22443, AluJb-dup7260, L2b-dup26840, AluSx1-dup30337, L2a-dup163990, L2a-dup163989, AluSq2-dup15156, L1PA15-dup2196, AluJb-dup36574

AluJr-dup82212, AluSq2-dup8706, AluSg7-dup5445, AluSx-dup32527, AluSz6-dup44790, AluSx3-dup20134, AluSx1-dup84605, AluSc-dup32567, HAL1-dup18700, AluSx1-dup56311

AluSc8-dup6033, AluY-dup27178, AluSx1-dup27642, AluYa5-dup575, L1MA9-dup11533, AluSx1-dup71941, AluSq2-dup38744, AluSz-dup65021, AluSz-dup18708, AluSc5-dup6328


PC_2
20
10
0
-10
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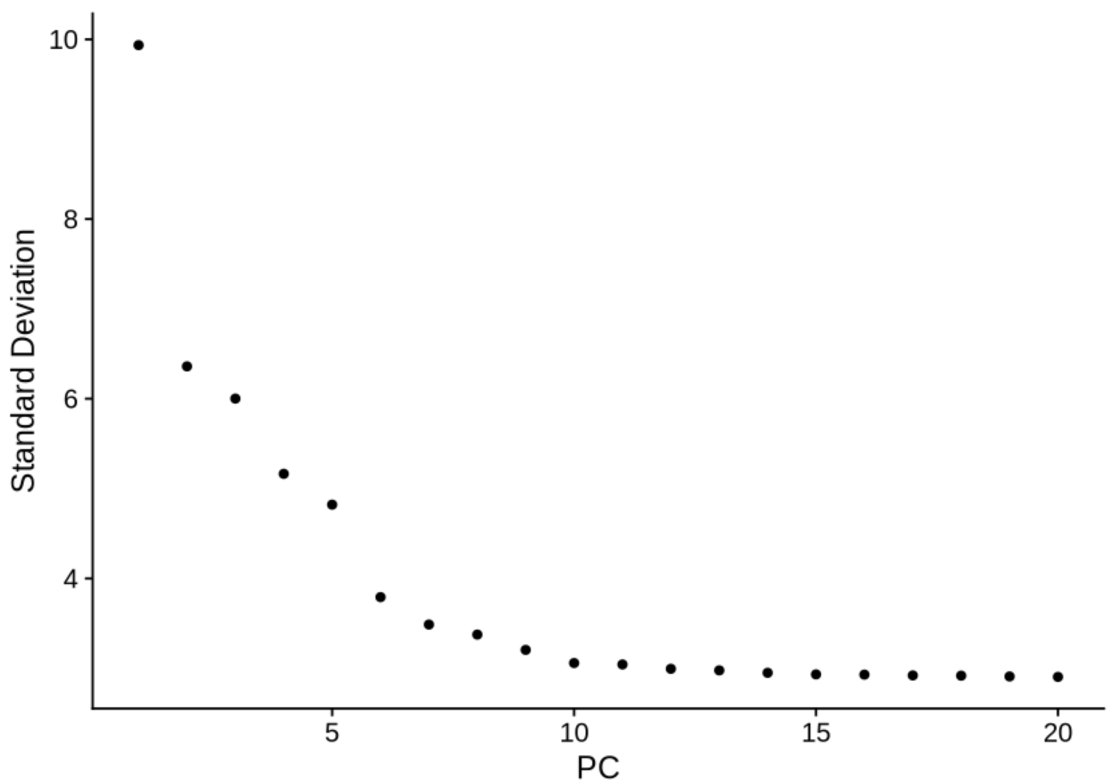
10
20
PC_1


Standard Deviation
10
8
6
4
5
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15
20
PC

[11]:
```
objTE_stellarscope <- FindNeighbors(objTE_stellarscope, dims = 1:12, k.param =
↪20)
objTE_stellarscope <- FindClusters(objTE_stellarscope, resolution = 1.2)
objTE_stellarscope <- RunUMAP(objTE_stellarscope, dims = 1:12)
DimPlot(objTE_stellarscope, reduction = "umap")
```

```
Computing nearest neighbor graph


Computing SNN


Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1056
Number of edges: 36086

Running Louvain algorithm…
Maximum modularity in 10 random starts: 0.7905
Number of communities: 8
Elapsed time: 0 seconds

Warning message:
"The default method for RunUMAP has changed from calling Python UMAP via
reticulate to the R-native UWOT using the cosine metric
To use Python UMAP via reticulate, set umap.method to 'umap-learn' and metric to
'correlation'
This message will be shown once per session"
15:05:20 UMAP embedding parameters a = 0.9922 b = 1.112

15:05:20 Read 1056 rows and found 12 numeric columns

15:05:20 Using Annoy for neighbor search, n_neighbors = 30

15:05:20 Building Annoy index with metric = cosine, n_trees = 50

0%   10   20   30   40   50   60   70   80   90   100%

[----|----|----|----|----|----|----|----|----|----|

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15:05:20 Writing NN index file to temp file
/mnt/TEdataStorage_2T/tmp/Rtmpu53vHH/filea91a552b5cbd6
```
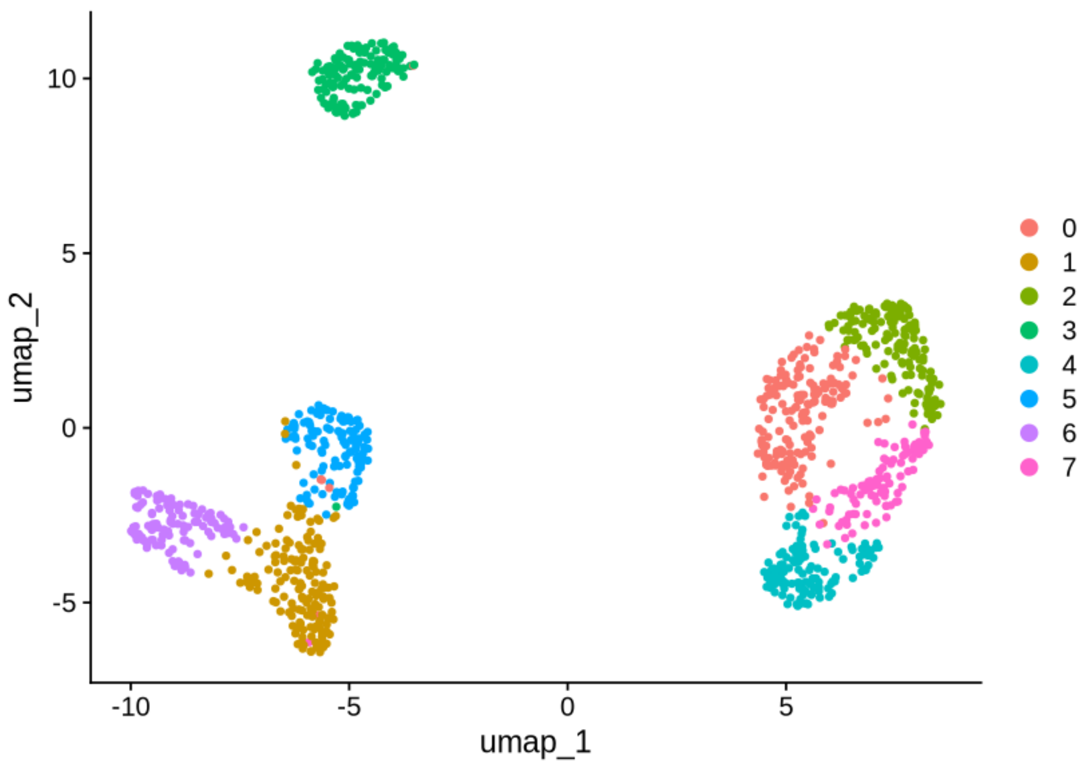
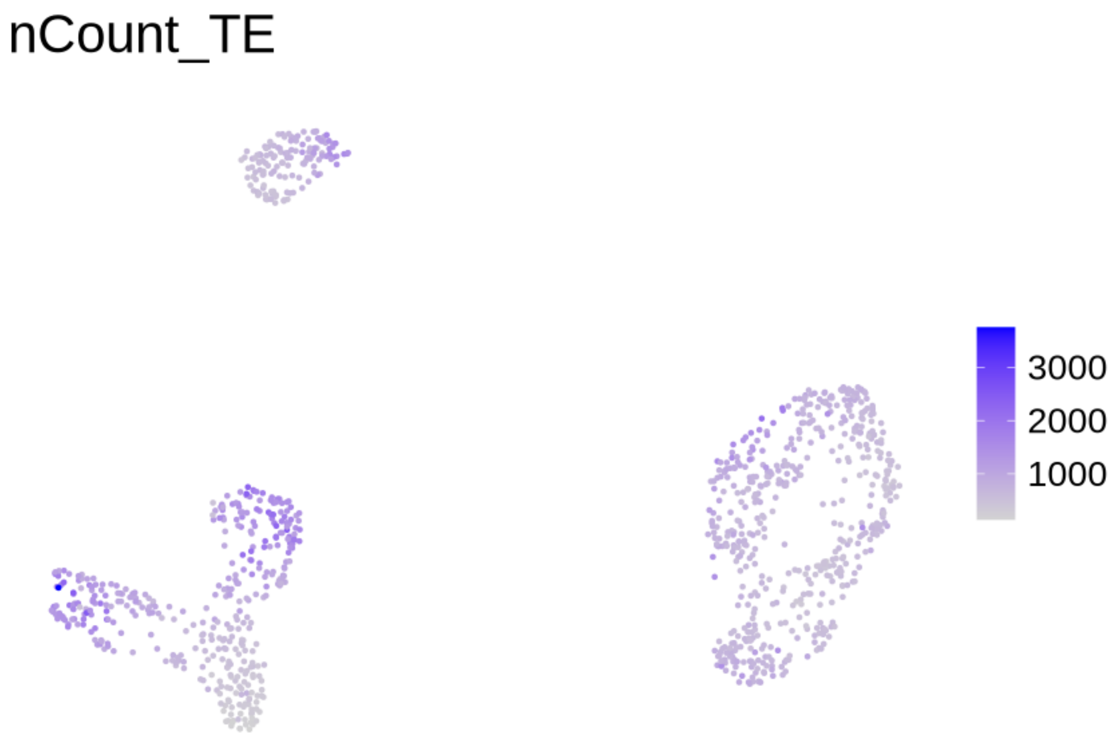

```
15:05:20 Searching Annoy index using 1 thread, search_k = 3000

15:05:20 Annoy recall = 100%

15:05:22 Commencing smooth kNN distance calibration using 1 thread
 with target n_neighbors = 30

15:05:25 Initializing from normalized Laplacian + noise (using RSpectra)

15:05:26 Commencing optimization for 500 epochs, with 40400 positive edges

15:05:26 Using rng type: pcg

15:05:29 Optimization finished
```




```
[12]: FeaturePlot(objTE_stellarscope, reduction = "umap", features = "nCount_TE", pt.
      ↪size = 0.5) +
        theme_void() +
        theme(text=element_text(size=20))
```



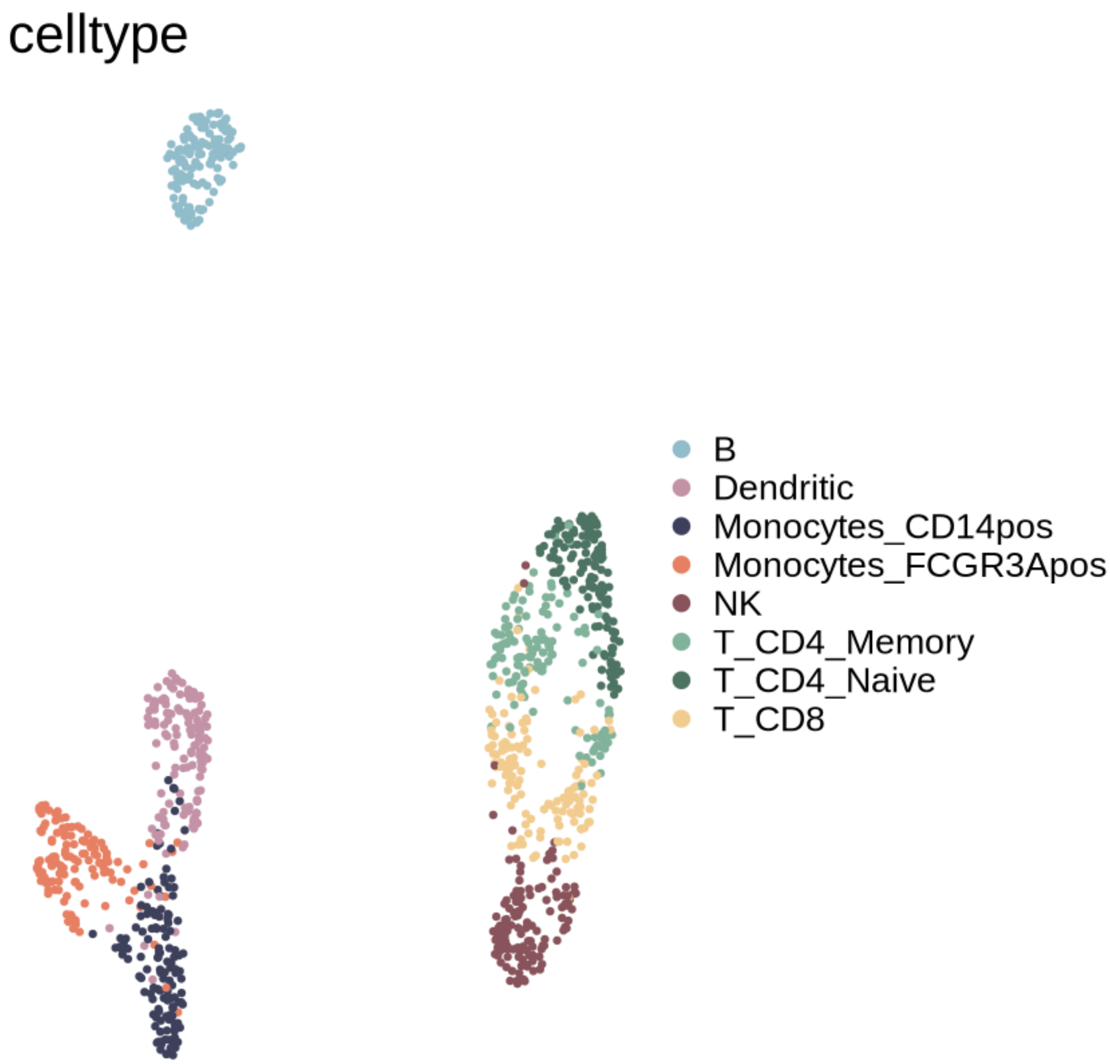
[13]:
```
# celltypes <- read.table("/mnt/TEdata/TEbenchmarking/data/whitelists/
↪celltype_annotation_sub_min.tsv")
# table(Cells(objTE_stellarscope) %in% celltypes$V1)

objTE_stellarscope$celltype <- celltypes$V2[match(Cells(objTE_stellarscope),␣
↪celltypes$V1)]
table(objTE_stellarscope$celltype)
```

```
                B         Dendritic   Monocytes_CD14pos Monocytes_FCGR3Apos
              132               132                 132                 132
               NK       T_CD4_Memory        T_CD4_Naive               T_CD8
              132               132                 132                 132
```

[14]:
```
options(repr.plot.width=7, repr.plot.height=7)

DimPlot(objTE_stellarscope, reduction = "umap", group.by = "celltype",
        cols=PBMCcelltypeColors,
        shuffle=T, pt.size = 1) +
  theme_void() +
  theme(text=element_text(size=20))
```

```
ggsave(paste0("figures_",dataset_id,"/
↪umap_TEs_locus_celltypes_stellarscope_",mode,"_noplatelet.png"), device =␣
↪"png")
ggsave(paste0("figures_",dataset_id,"/
↪umap_TEs_locus_celltypes_stellarscope_",mode,"_noplatelet.pdf"), device =␣
↪"pdf")
```

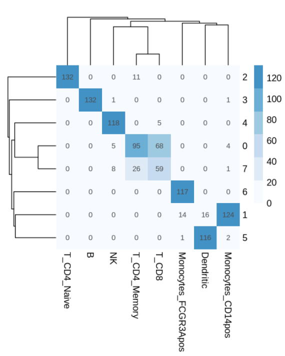
```
Saving 7 x 7 in image
Saving 7 x 7 in image
```

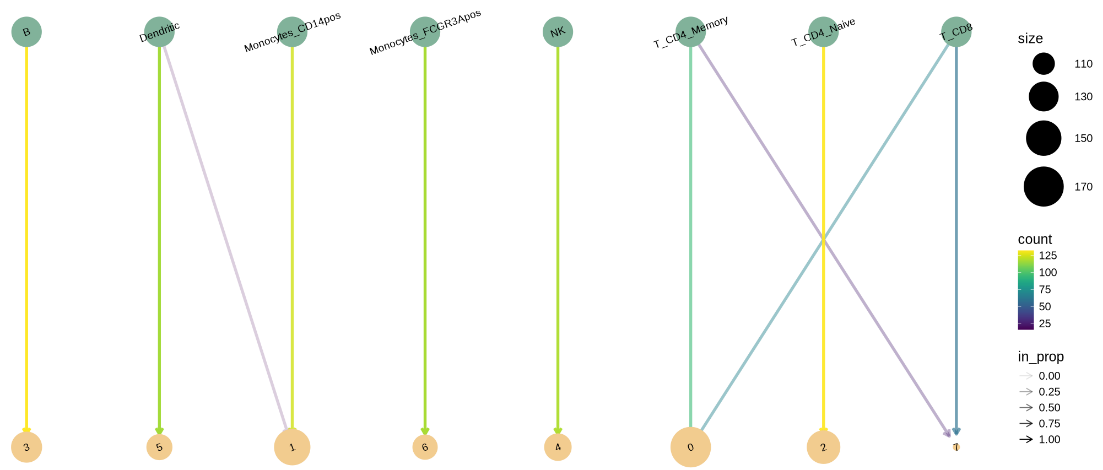


[15]:
```
saveRDS(objTE_stellarscope, paste0("data_", dataset_id, "/stellarscope_",␣
↪dataset_id, "_", mode, "seuratObj_noplatelet.RDS"))
```

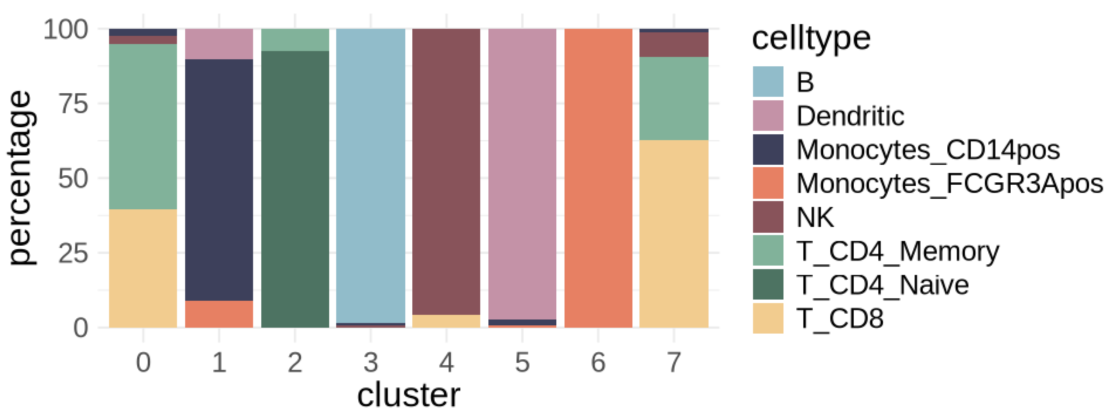
```
[16]: options(repr.plot.width=17, repr.plot.height=7)

# Add cluster information from objTE_clustered to objGenes_clustered
objTE_stellarscope$clusters.0.2 <- objTE_stellarscope$seurat_clusters
objTE_stellarscope$clusters.0.1 <- objTE_stellarscope$celltype

concordanceTable <- table(objTE_stellarscope$seurat_clusters, objTE_stellarscope$celltype)



pheatmap(concordanceTable, display_numbers = T, color = brewer.pal(9,'Blues')[1:6],
         border_color = NA, number_format = "%.0f", cellwidth = 25, cellheight = 25)

# Generate the clustree plot
clustree(objTE_stellarscope, prefix = "clusters.", node_text_angle=20, node_text_size=4) + theme(text=element_text(size=15)) +
  scale_color_manual(values=alpha(c("#81B29A","#F2CC8F"), 0.6)) + scale_size(range = c(3,20)) +
  guides(colour = FALSE)
```

```
Scale for size is already present.
Adding another scale for size, which will replace the existing scale.
Warning message:
"The `<scale>` argument of `guides()` cannot be `FALSE`. Use "none"
instead as
of ggplot2 3.3.4."
```





[17]:
```
options(repr.plot.width=8, repr.plot.height=3)

df <- melt(concordanceTable)
df <- df[df$value!=0,]
colnames(df) <- c("cluster","celltype","nCells")
df$cluster <- as.character(df$cluster)
df$clusterSize <- table(objTE_stellarscope$seurat_clusters)[df$cluster]
df$percentage <- as.numeric(df$nCells / df$clusterSize *100)

df$cluster <- factor(df$cluster, levels=c(0,1:length(unique(df$cluster))))


ggplot(df, aes(x=cluster, y=percentage, fill=celltype)) +
  geom_col() +
  scale_fill_manual(values = PBMCcelltypeColors)+
  theme_minimal() + theme(text=element_text(size=18))
ggsave(paste0("figures_10xPBMC/
↪clusterIdentity_barplot_Stellarscope,",mode,"_noplatelet.pdf"), width=8,␣
↪height=3)
```



```
[ ]:
```